

Block D

Replication & robustness as research questions

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MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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Papers and materials

Harvey, C. R., Liu, Y., & Zhu, H. (2016). *...and the Cross-Section of Expected Returns*. *RFS*, 29(1), 5–68.

Jensen, T. I., Kelly, B., & Pedersen, L. H. (2023). *Is There a Replication Crisis in Finance?* *JF*, 78(5), 2465–2518.

Kelly, B., Malamud, S., & Zhou, K. (2024). *The Virtue of Complexity in Return Prediction*. *JF*, 79(1), 459–503.

Nagel, S. (2025). *Seemingly Virtuous Complexity in Return Prediction*. NBER WP 34104.

Live: `notebooks/06_kmz_nagel.ipynb`

Back to the opener's table

Every row today was a **claim**:

- nonlinear interactions carry risk-premium information (Gu)
- past prices predict returns (Krauss)
- advanced classifiers beat LR (Lessmann)
- no-arbitrage structure improves pricing (CPZ)

What would it take to replicate one of them? And what counts as success?

Paper card: What counts as replication?

	Harvey, Liu & Zhu (2016), <i>RFS</i> 29(1)	Jensen, Kelly & Pedersen (2023), <i>JF</i> 78(5)
Question	After hundreds of tested factors, what t-statistic should a new factor clear?	Is there a replication crisis in finance?
Data	316 published factors.	153 factors in 93 countries; US data from 1926.
Method	Multiple-testing adjustments (Bonferroni, Holm, BHY).	Replication = a significant CAPM alpha in the original direction; a Bayesian hierarchical model shrinks each factor toward its cluster.
Main result	A new factor should clear $t > 3.0$, not 2.0.	The replication rate (→ poll in class).
Watch for	The hurdle depends on how many hypotheses the field has already tried.	What counts as "replicated"? Fix the definition before you test.
Code & data	—	Factors at jkpfactors.com ; code on GitHub (bkelly-lab/ReplicationCrisis).

Harvey, Liu & Zhu (2016)

- They catalogue **316** factors
- After that much testing, $t > 2$ is not a meaningful hurdle
- Proposal: $t > 3.0$ for new factors, using multiple-testing adjustments

The hurdle depends on how many hypotheses the field has already tried.

Jensen, Kelly & Pedersen (2023): design

- **153 factors, 93 countries**; US data from 1926
- **Replication** = a significant **CAPM alpha** in the original direction
- A Bayesian hierarchical model (empirical Bayes): each factor shrinks toward its cluster, using all factors jointly
- Data: jkpfactors.com · Code: github.com/bkelly-lab/ReplicationCrisis

Jensen, Kelly & Pedersen (2023): results

Definition	US replication rate
JKP, Bayesian	82.4%
JKP, Benjamini–Yekutieli (critical $t \approx 2.7$)	75.6%
Hou, Xue & Zhang (2020), their definitions	35%

13 themes; most matter in the tangency portfolio.

Why 82% here and 35% there? (3 min)

List the choices that differ:

- raw return vs **CAPM alpha**
- treatment of **microcaps** and the weighting
- the **multiple-testing** correction

The replication rate is a function of the definition.

Choose the definition **before** you test. That is hypothesis development.

Paper card: Kelly, Malamud & Zhou (2024)

	<i>Journal of Finance</i> 79(1): 459–503
Question	Can a model with far more parameters than observations time the stock market better out of sample?
Data	Monthly US market excess return, 1926–2020; 15 predictors (14 Welch–Goyal variables plus the lagged market return).
Method	Random Fourier features of the predictors (up to $P = 12,000$) in ridge regression on a rolling 12-month window, averaged over 1,000 random draws. Theory shows why complexity can help.
How it is tested	An out-of-sample timing strategy (forecast $\times$ return) over 1,091 months: Sharpe ratio, alpha against the market and R^2_{OOS} , plotted against complexity P/T .
Main result	The "virtue of complexity": timing performance rises with P/T , even when R^2_{OOS} is negative. Information ratio about 0.3, alpha $t = 2.6$ – 2.9 .
Watch for	With 12 observations and 12,000 parameters, what can the model actually learn? Returns are scaled by their trailing volatility.
Code & data	Welch–Goyal data are public; community replications exist on GitHub.

Kelly, Malamud & Zhou (2024): the claim

Theory: out-of-sample timing performance can *rise* with complexity P/T , even when $R_{\text{OOS}}^2 < 0$.

Empirics:

- target: monthly market excess return
- $K = 15$ predictors: 14 Welch–Goyal variables + the lagged market return
- random Fourier features, with $\omega_i \sim N(0, I)$ and $\gamma = 2$:

$$z_t = \sqrt{2/P} \left[\cos(\gamma \omega'_i x_t), \sin(\gamma \omega'_i x_t) \right]_{i=1}^{P/2}$$

- P up to 12,000 · **rolling** $T = 12$ · ridge $z \in \{10^{-3}, \dots, 10^3\}$ · returns scaled by trailing volatility

Live 1: the "virtue of complexity" curve

06_kmz_nagel.ipynb §1–2 (precomputed)

- timing strategy return: $\hat{r}_{t+1} \times r_{t+1}$
- plot the annualized Sharpe ratio and R^2_{OOS} against P/T , one line per z

Ours (ridgeless): the Sharpe ratio rises from -0.10 to 0.37 . The dip at $P/T = 1$ is the interpolation threshold, which KMZ's theory predicts.

Now: what is the model actually doing?

Paper card: Nagel (2025)

	NBER Working Paper 34104
Question	What does KMZ's high-complexity model actually do with a 12-month window?
Data	KMZ's replication data: the market return and the same 15 predictors.
Method	Shows that the ridgeless random-feature forecast is close to a Gaussian kernel regression, i.e. a weighted average of the last 12 returns. Builds a simple volatility-timed momentum benchmark; runs a placebo with simulated reversals and a bootstrap of the predictors.
How it is tested	Alphas and information ratios against the market; spanning regressions of the KMZ strategy on the kernel and momentum strategies.
Main result	Information ratios: KMZ strategy 0.255, kernel 0.306, vol-timed momentum 0.388. Given the kernel strategy, the KMZ alpha is zero. Placebo: → poll in class.
Watch for	The numbers replicate; the interpretation changes. Did Nagel refute the theory or its empirical reading?
Code & data	Public data; our notebook 06 rebuilds the main tests.

Nagel (2025): the reinterpretation

With $T = 12$ and $P \gg T$, the ridgeless RFF forecast $\approx$ **Gaussian kernel regression**:

$$\hat{r}_{t+1} \approx k(x_t, X) K(X, X)^{-1} r, \quad k(x, x') = e^{-\frac{\gamma^2}{2} \|x - x'\|^2}$$

This is a weighted average of the **last 12 returns**, with weights set by how similar today's predictors are to each past month's.

In effect it is a **volatility-timed momentum** strategy that happened to do well.

Nagel also drops KMZ's standardization of returns by trailing volatility: on its own, that step can create spurious predictability (his Appendix A).

Live 2: Nagel's evidence ($T = 12, P = 12,000$)

Test (1930–2020)	Nagel	Ours
Correlation, RFF vs kernel forecast	0.93	0.97
IR: RFF / kernel / vol-timed momentum	0.255 / 0.306 / 0.388	0.283 / 0.258 / 0.384
RFF alpha t , given the kernel strategy	−0.12	1.33
RFF alpha t , given vol-timed momentum	0.95	1.32
Bootstrapped predictors: IR	0.228 (vs 0.255)	not run

Vol-timed momentum replicates almost exactly and spans the RFF strategy. The kernel spans it less well in our data (t falls from 1.88 to 1.33 as the RFF draws rise from 20 to 100; Nagel uses 1,000).

Live 3: placebo

Add noise with **reversals** to the returns, $\tilde{r}_t = r_t + \xi_t - 0.2 \xi_{t-1} - 0.2 \xi_{t-2}$, keep the other 14 predictors, and re-run.

→ The RFF strategy **loses money**: Nagel $t = -1.84$; ours $t = -2.02$, and 19 of 20 simulated paths lose.

It does not learn the sign of the dynamics. It **imposes** momentum.

After publication: 2021–2025

IR, 60 months	RFF	Kernel	Vol-timed momentum
Ours	−0.20	−0.19	0.02

None of the strategies has earned anything since the sample ended.

But 60 months have little power: even if the true IR were 0.28, the expected t would be about $0.28 \times \sqrt{5} \approx 0.6$.

Is this evidence against KMZ, or no evidence at all?

Discussion (5 min)

1. Did Nagel refute KMZ's **theory** or their **empirical interpretation**?
2. Which hypothesis did KMZ's empirical test actually test?
3. Which single diagnostic would you add to **your** paper to pre-empt a critique like this?

Takeaway

A replication is a hypothesis test about the paper.

Fix what counts as success before you run it.

Homework: re-run 06 with $T = 60$ and $T = 120$. Does the momentum reading survive?

Next: Block E, replication from the author's side